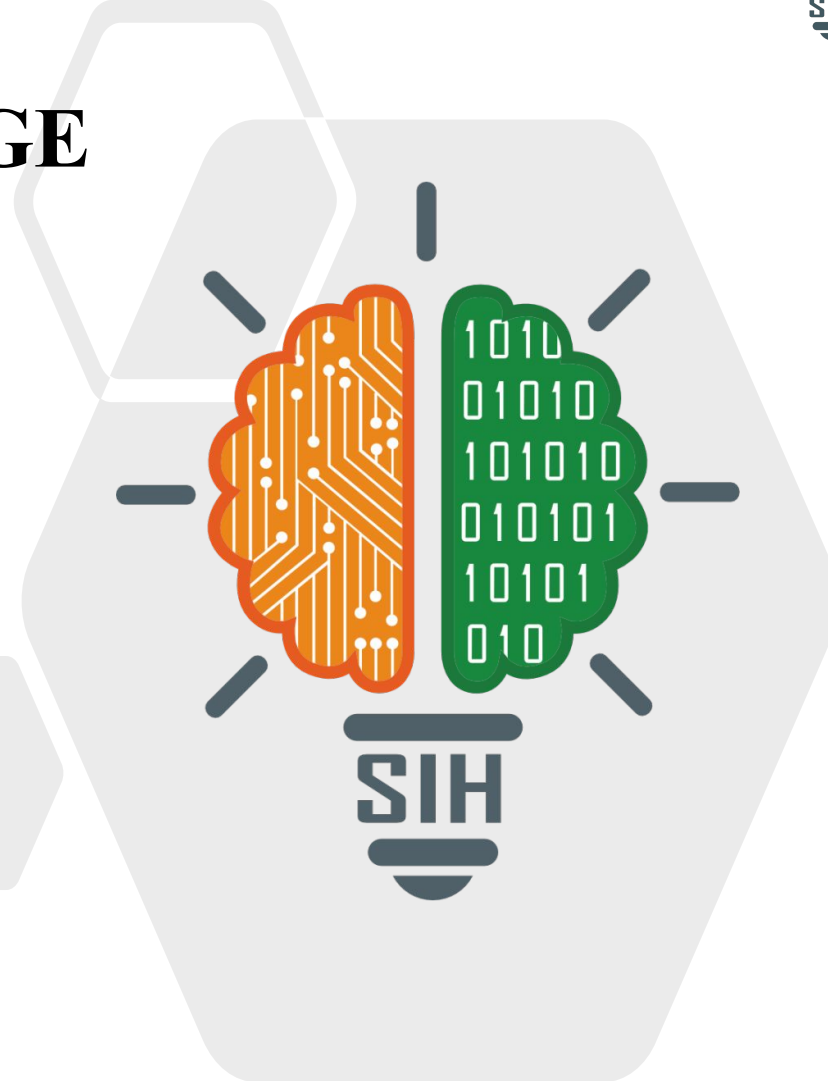


TITLE PAGE

- **Problem Statement ID – SIH26022**
- **Problem Statement Title- Design and develop drying and compact packaging system to support home-based agarbatti manufacturing by rural women artisans**
- **Theme- Agriculture, Foodtech & Rural Development**
- **PS Category- Hardware**
- **Team ID-**
- **Team Name - Team Titan**



AGARSMART - Smart Drying & Quality Management System for Rural Agarbatti Artisans

PROPOSED SOLUTION



Smart Solar Drying Chamber - Provide enclosed, controlled and hygienic drying using solar power



Temperature & Humidity Control - Sensor monitor the chamber and automatically control heating for uniform drying



Camera-Based Quality check - A camera/colour sensor checks consistency and visible defects and give grade



Moisture-Resistant Packaging & Sealing - A low-cost sealing unit provides protection against moisture and help preserve fragrance

PROBLEM ADDRESSED



Weather-dependent drying - Traditional drying depends on sunlight and weather

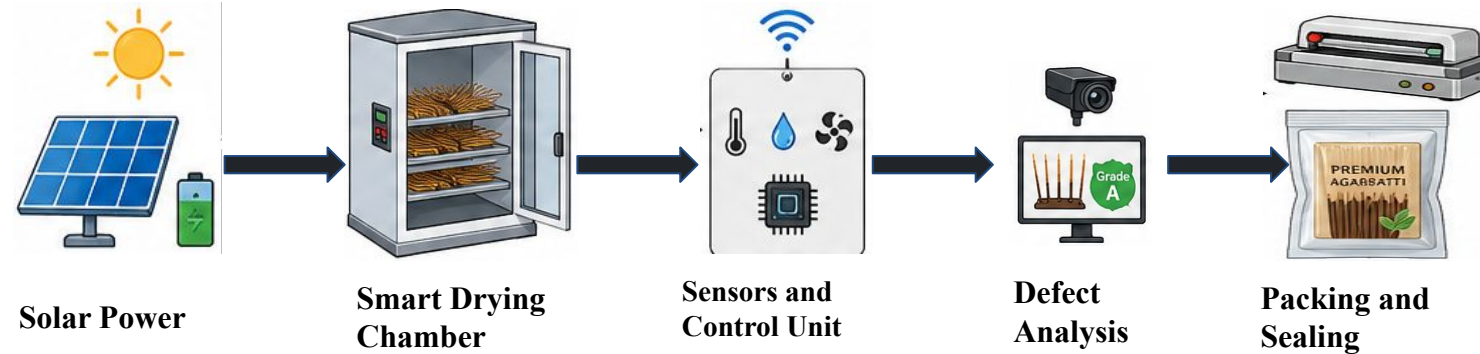


Fragrance & quality loss - Improper drying conditions can affect fragrance, colour and overall quality



Moisture-sensitive packaging - Poor packaging can allow moisture entry and reduce fragrance and shelf life

PROCESS FLOW



NOVELTY



Smart Quality Monitor - Monitors temperature, humidity, and drying time to assess drying quality



Fragrance Protection - Maintain safe drying conditions to minimize fragrance loss



Smart Drying Control - Automatically regulates heating and airflow for uniform drying



Batch Quality Record - Stores drying data and quality history for each batch



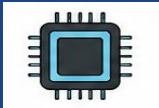
Predictive Maintenance AI - Detects possible faults before equipments failure

SOFTWARE WORKFLOW



Data Acquisition

Read Temperature
Read Humidity
Capture Image
Get System Status



Data Processing

Filter & Validate
Noise Reduction
Feature Extraction



Decision Making

Evaluate drying
conditions
Determine control
action
Quality Assessment



Actuator Control

Control fan speed
Control heater
Adjust Ventilation
Maintain optimal
condition



Data Logging & Storage

Store sensor data
Store quality grade
Store batch record
Timestamp & ID



User Interface & Alerts

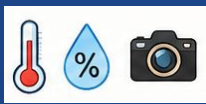
Real-time
Dashboard
Alerts &
Notification
Batch history

HARDWARE WORKFLOW



Solar Power Supply

Solar Panel
Charge Controller
Battery Backup
DC Regulation



Sensors

DHT22(Temp &
Humidity)
Camera/Colour
Sensor
Airflow Sensor



Microcontroller (ESP32)

Data acquisition
Processing
Decision Making
Communication



Actuators

DC Fan
Heater
Vent/Servo
Sealing Unit
Control



Display & Communication

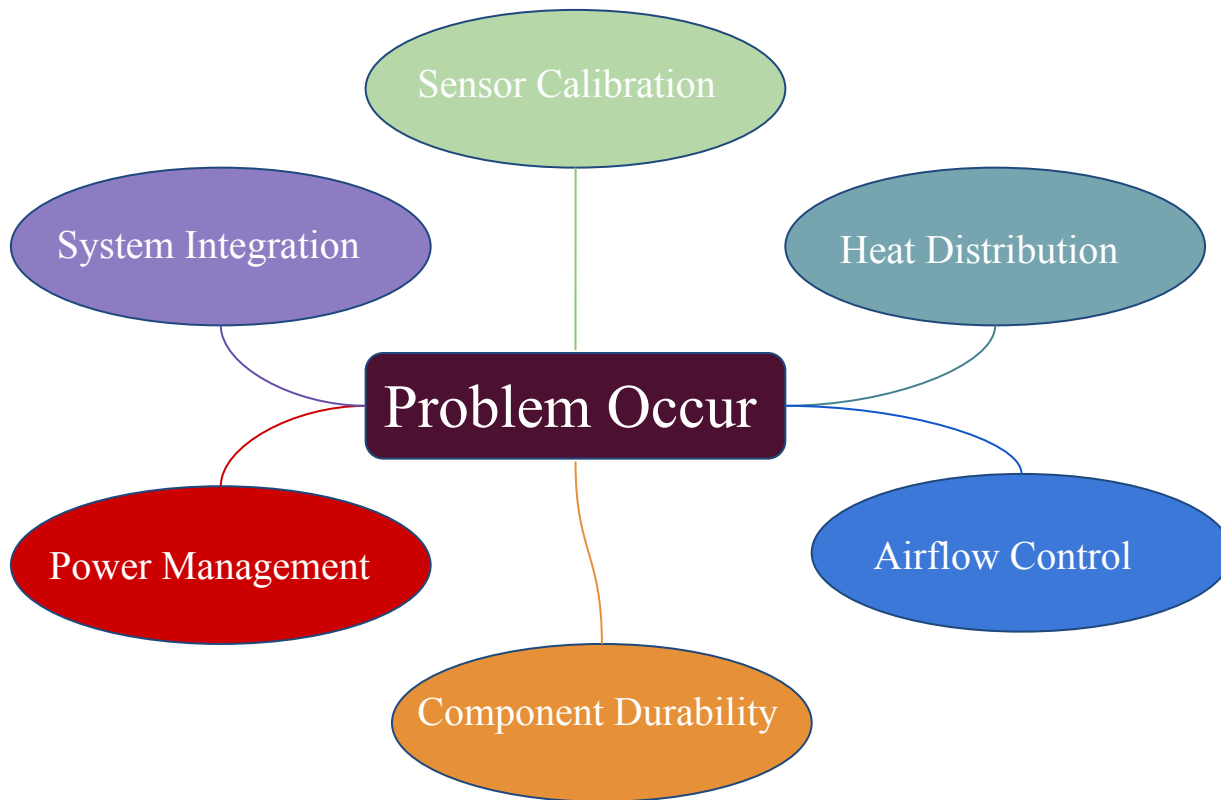
LCD/OLED display
Wifi module(IOT)
Cloud Dashboard



Packing Unit

Heat Sealing Unit
Moisture resistant
Pouch

Problem & Challenges Faced



Solution for the Problem faced

Calibrated Sensors - Use calibrated temperature and humidity sensors for reliable readings

Insulated Chamber - Use thermal insulation to maintain stable internal temperature

Optimized Airflow - Position DC fans strategically for even air circulation

Protected Electronics - Isolate the control unit from heat and moisture exposure

Smart Power Control - Prioritize solar power and use battery backup for continuous operation

Modular Control System - Use a centralized microcontroller for reliable sensor-fan-heater coordination

IMPACT



SOCIAL

Supports rural women and SHGs with easy and reliable agarbatti production



ECONOMIC

Reduce drying losses and product wastage
Improve the product quality and market value



ENVIRONMENTAL

Uses solar energy to reduce grid electricity consumption
Supports eco-friendly packaging



TECHNICAL

Enables automated temp, humidity, and airflow control
Provides smart monitoring



LIVELIHOOD

Enable all weather production and more consistent output
Helps artisans increase productivity and earnings

BENEFITS

Uniform and Hygienic drying

Lower energy Dependence

Portable and Rural-friendly

Better fragrance retention

Moisture-resistant packaging

Scalable for household and SHG production

Research & References

1. Solar Drying Technology

Solar-powered drying for controlled and uniform drying.

Reduces dependence on open-sun drying.

2. IoT-Based Monitoring

ESP32-based temperature and humidity monitoring.

Real-time drying and battery status tracking.

3. Smart Drying Control

Sensor-based automatic fan and heater control.

Maintains controlled drying conditions.

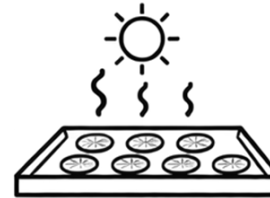
4. Aroma-Preserving Packaging

Moisture-resistant barrier packaging for dried agarbattis.

Solar Agarbatti Drying Research Paper

https://docs.google.com/document/d/1JsWojOf_DrxEq-FagyMB7ZXfJP-Jr9pt/edit?usp=sharing&ouid=117108260815598326645&rtpof=true&sd=true

MARKET / TECHNOLOGY BENCHMARKING

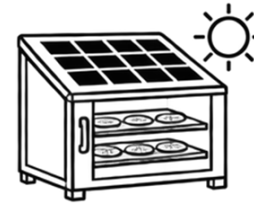


1 Traditional Open-Sun Drying

What it does: Low-cost natural drying.

Limitations: Weather-dependent and uneven.

Why our system: Controlled, enclosed, all-weather drying.



2 Conventional Solar Dryers

What they do: Use solar energy for drying.

Limitations: Limited automated control.

Why our system: Solar + sensors + automatic control.



3 Smart Drying Systems

What they do: Monitor drying parameters.

Why our system: Combines monitoring, automatic control, and packaging in one low-cost system.